Madinah Operations Intelligence

From municipal reports to proactive decisions

Date Range: 05/10/25 – 05/04/26

District: All

Category: All

Shift: All

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MADINAH OPERATIONS INTELLIGENCE

- Overview
- Operations
- AI Insights
- Project Info

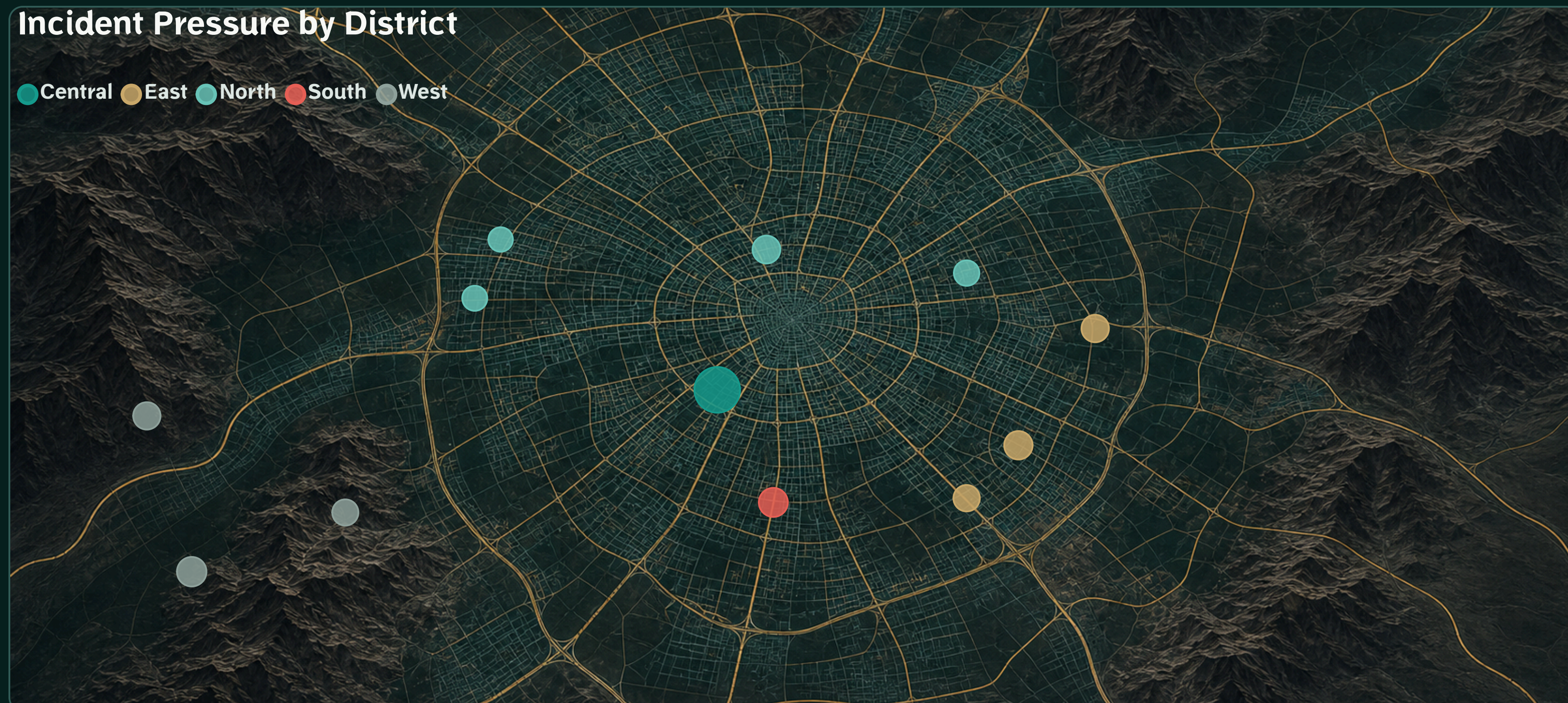
30K Incoming Reports

73.5 % SLA Compliance

28.3 Resolution Hours

8.4 % Repeat Rate

Incident Pressure by District



Operational Insight

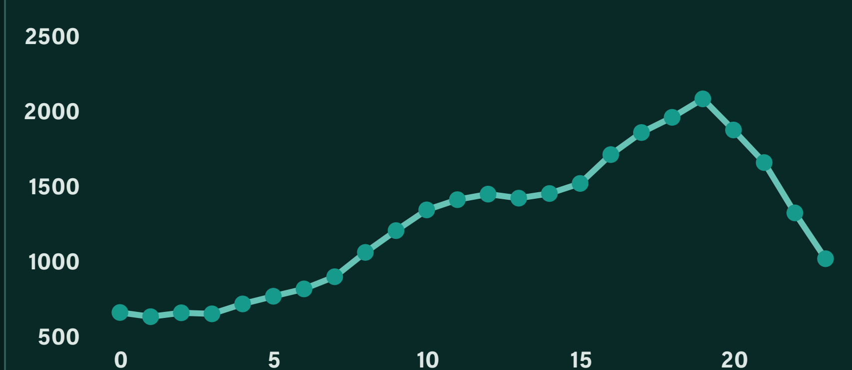
Finding
35.7% of reports in All Districts occur during the evening period.

Repeat rate: 8.4%
Operational pressure: Moderate

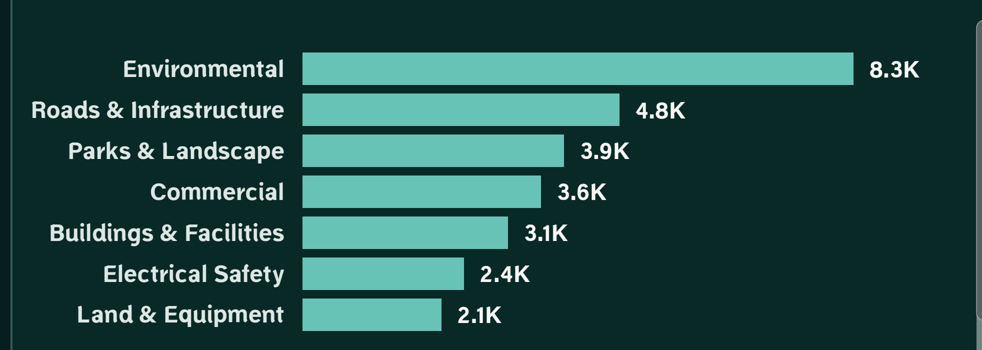
Suggested action
Review evening field-team allocation and prioritize repeat-report locations.

AI-assisted

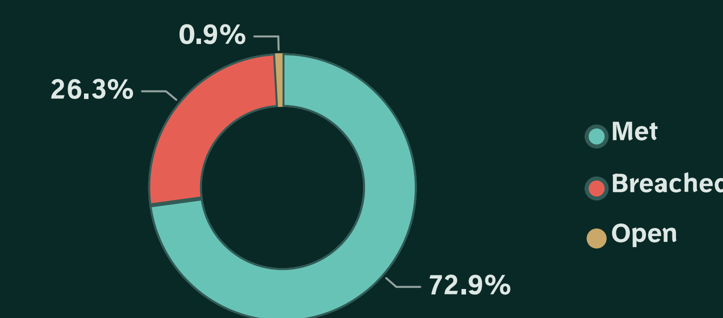
Incident Demand by Hour



Reports by Service Category



SLA Status



PORTFOLIO PROJECT

Synthetic data

Inspired by internship experience

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PORTFOLIO PROJECT
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Operations & SLA Performance

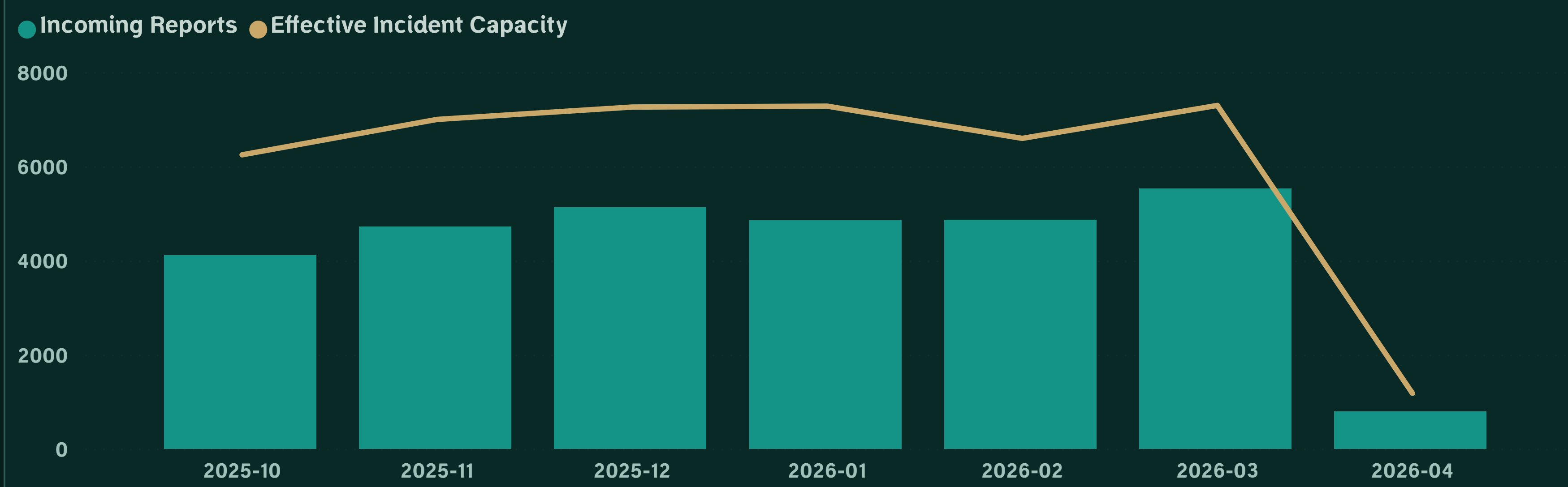
Capacity, workload and service-level monitoring

Date Range: 05/10/25 – 05/04/26
District: All
Shift: All

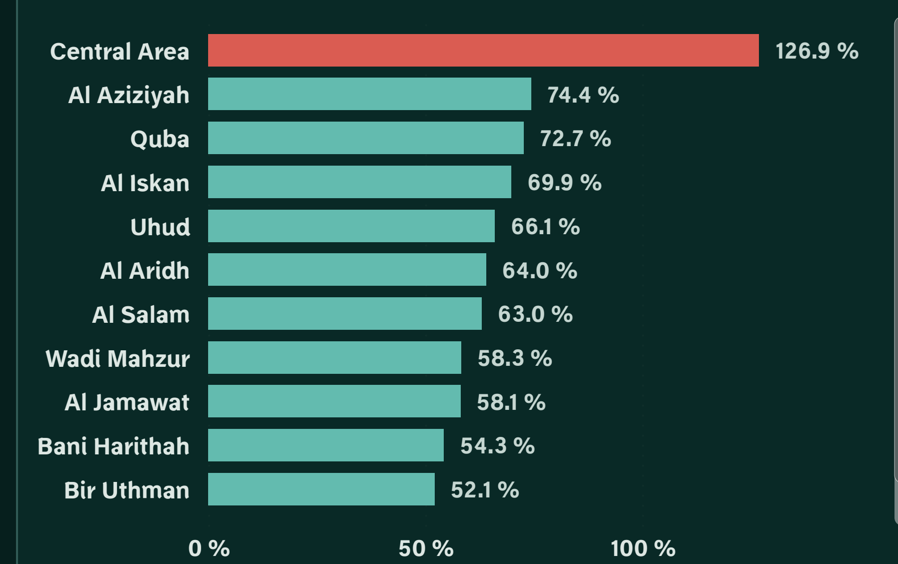
42.84K	Effective Capacity
70.0 %	Capacity Utilization
97.3 %	Team Availability Rate
267	Open Reports

Demand vs Effective Capacity

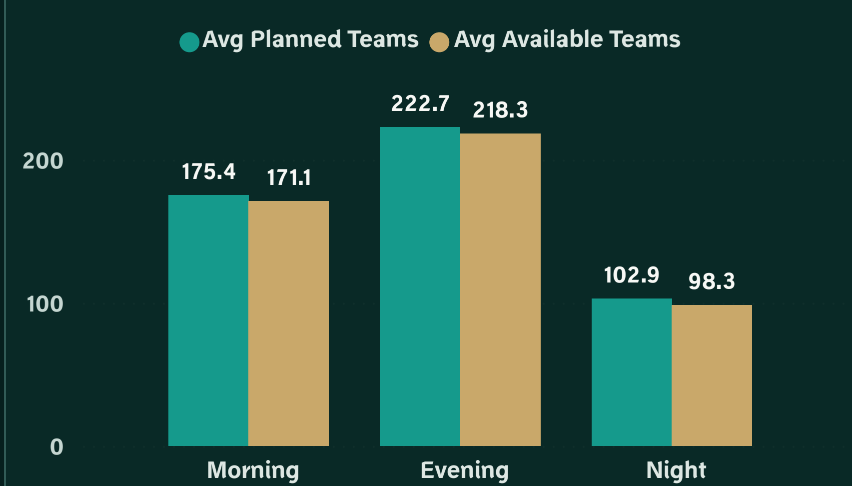
Monthly totals • Apr 2026 includes data through 5 Apr



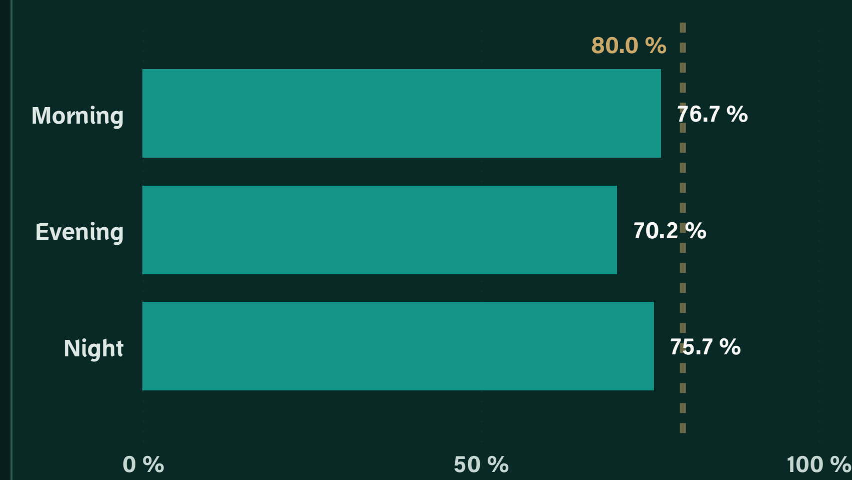
Capacity Utilization by District



Average Teams by Shift



SLA Compliance by Shift



District Operational Priorities

District	Utilization	SLA	Open
Uhud	66.1 %	77.8 %	21
Quba	72.7 %	77.6 %	22
Central Area	126.9 %	59.3 %	63
Al Iskan	69.9 %	78.7 %	23
Al Aziziyah	74.4 %	78.1 %	29
Al Aridh	64.0 %	77.3 %	12
Total	83.0 %	71.2 %	170

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OPERATIONS INTELLIGENCE

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AI-Assisted Operational Insights

Risk detection, demand signals and explainable recommendations

Date Range 05/10/25 05/04/26

District All

Shift All

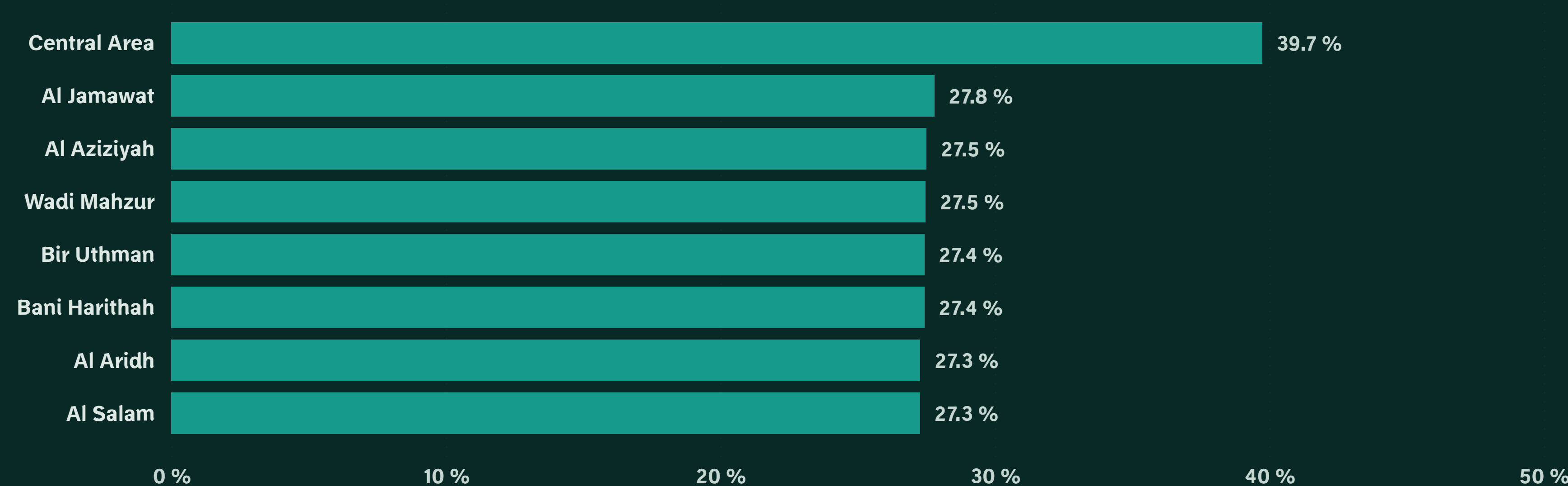
30.1 % Aveg. Predicted Risk

4K High Risk Reports

53 Open High Risk

76.6 % Model Recall

Predicted SLA Risk by District



Explainable Recommendation

AI Recommendation

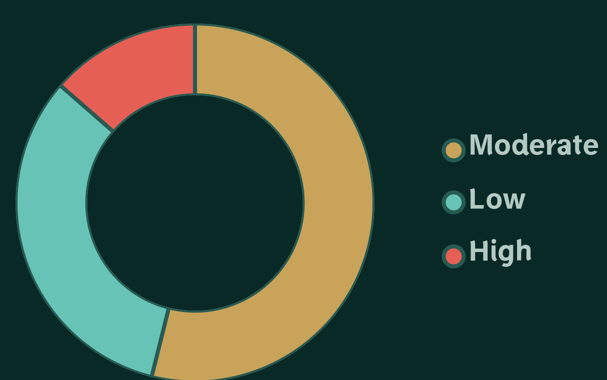
Priority district
Central Area

Predicted SLA breach risk: 39.7%
Open high-risk reports: 45
Capacity utilization: 126.9%
Current SLA compliance: 59.3%

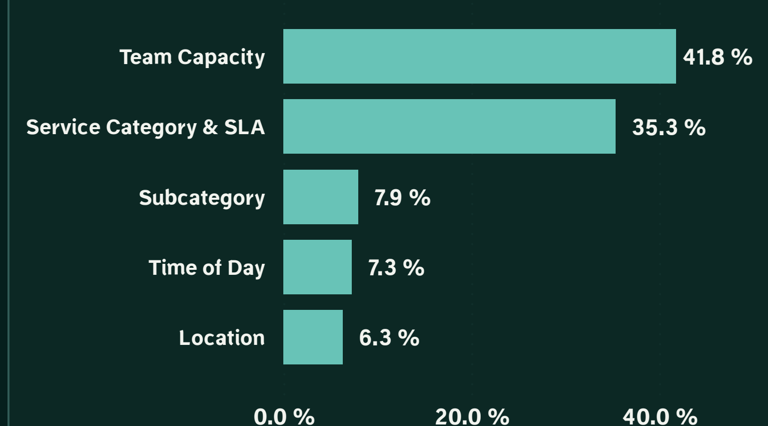
Recommended action
Add field capacity and prioritize model-flagged reports.

AI-assisted ▪ Human review required

Predicted Risk Distribution



Model Drivers



High-Risk Priority Queue

Incident ID	District	Risk	Priority	Shift
SYN-029957	Central Area	63.7 %	High	Evening
SYN-029818	Central Area	55.9 %	High	Evening
SYN-029621	Central Area	55.6 %	Low	Evening
SYN-029628	Central Area	55.6 %	Low	Evening
SYN-029973	Central Area	55.6 %	Low	Evening
SYN-029812	Central Area	55.2 %	Low	Evening
SYN-029645	Central Area	54.6 %	Medium	Evening
SYN-029650	Central Area	54.6 %	Medium	Evening
SYN-029813	Central Area	54.6 %	Medium	Evening
SYN-029819	Central Area	54.6 %	Medium	Evening
SYN-029937	Central Area	51.5 %	High	Evening

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Project Overview & Methodology

Business context, data architecture and AI approach

Project Summary

Madinah Operations Intelligence is an end-to-end BI and predictive analytics portfolio project inspired by internship experience at the Municipal Operations Monitoring Center. It transforms synthetic municipal report data into operational monitoring, capacity analysis, SLA risk prediction and explainable decision support.

SYNTHETIC DATA ▪ PRIVACY-SAFE ▪ PORTFOLIO PROTOTYPE

Project Facts

INTERNSHIP PERIOD
05 Oct 2025 – 05 Apr 2026

DOMAIN
Municipal Operations Intelligence

DATASET
30,000 synthetic municipal reports

CORE TOOLS
Power BI ▪ Python
DAX ▪ Power Query

FOCUS
BI ▪ Operations
Predictive AI

From Municipal Data to Proactive Decisions

01 SYNTHETIC DATA → 02 POWER QUERY & MODEL → 03 BI & DAX ANALYTICS → 04 AI RISK SCORING → 05 DECISION SUPPORT

Business Problem

- High report concentration near the Central Area
- Uneven capacity utilization across districts
- SLA breaches and repeat-report pressure
- Need for proactive prioritization

BUSINESS OUTCOME

Faster, evidence-based operational decisions

AI Methodology

Target: SLA breach before resolution

Models: sklearn Logistic Regression + AdaBoost

Selected: AdaBoost Decision Stumps

Time-based Train / Validation / Test split

Leakage-safe features available at report creation

AI-ASSISTED ▪ HUMAN REVIEW REQUIRED

Model Performance & Governance

TEST ROC-AUC	69.3%
MODEL RECALL	76.6%
DECISION THRESHOLD	27%

Synthetic data ▪ Six-month history

Portfolio prototype ▪ Not production-deployed

Predictions support—not replace—human decisions